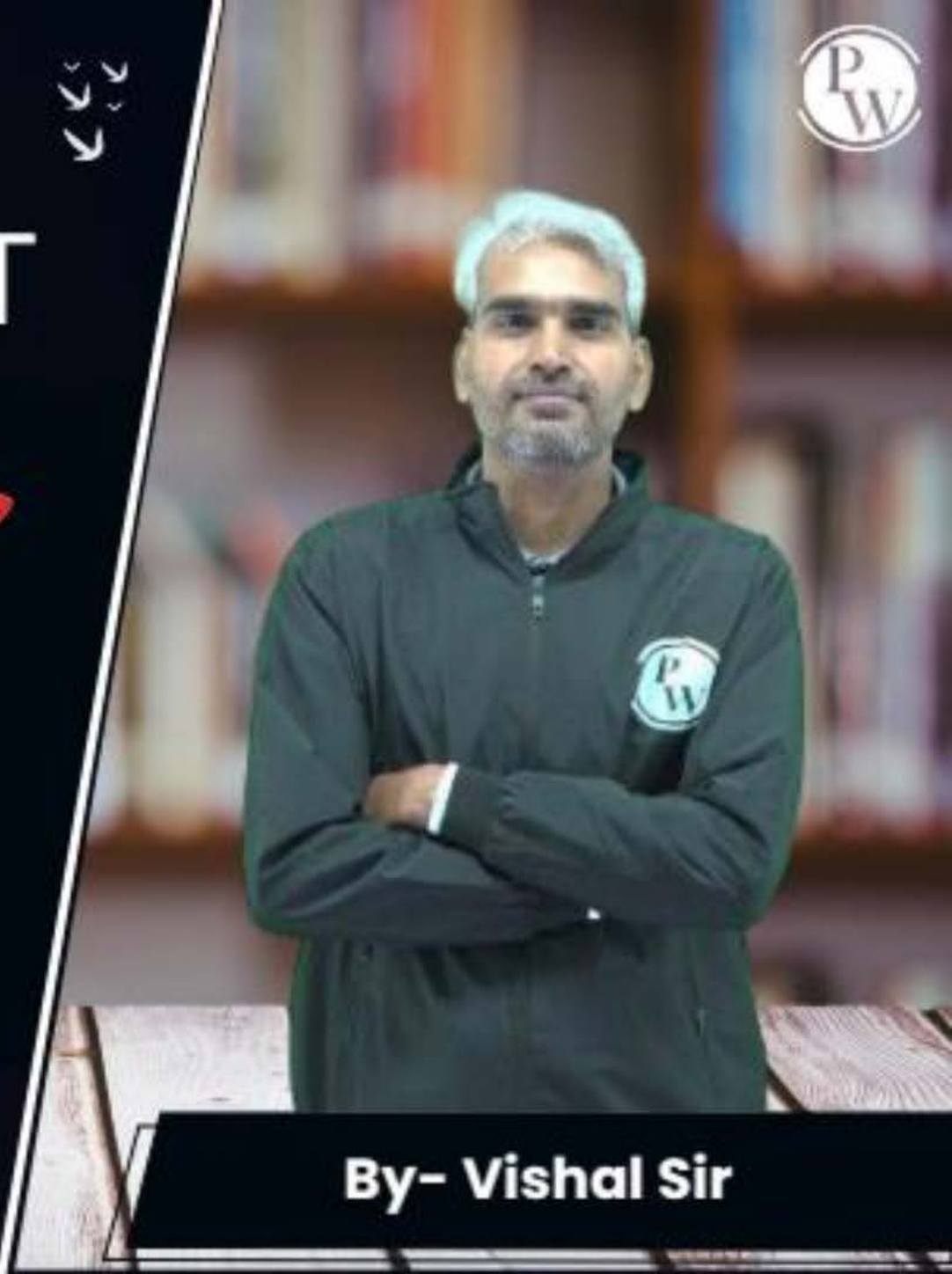
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 12



By- Vishal Sir

Recap of Previous Lecture



Topic — Number of partitions of a set

Topic — Number of equivalence relation on a set

Topic — Bell triangle

Slide

Topics to be Covered



- Topic: Partial order relation
- Topic: Partially ordered set (POSET)
- Topic: Comparability
- Topic: Total order relation and TOSET



Topic : Partial Order Relation

- A relation R on set A is called a Partial order relation if and only if relation R is Reflexive, anti-symmetric and transitive

Partial order = Reflexive + Anti-symmetric + Transitive

Slide

eg: Let $A = \{1, 2, 3\}$

$\Delta_A = \{(1,1), (2,2), (3,3)\}$

Reflexive ✓

Anti-symmetric ✓

Transitive ✓

∴ Partial order relation

Diagonal relation on set A is the smallest partial order relation on set A

* Diagonal relation on set A is the only relation which is an Equivalence relation as well as a Partial order relation

eg. Let $A = \{1, 2, 3\}$

$R = \{(1,1), (2,2), (3,3), (1,2), (1,3), (2,3)\}$

∴ Reflexive

∴ Anti-symmetric

∴ Transitive

Reflexive ✓
Anti-Symmetric ✓
Transitive ✓

} ∴ Partial order relation



Topic : Partially Ordered Set [POSET]

- Let R is a partial order relation on set A.
- Set A along with the partial order relation R defined on set A is called a POSET denoted by (A, R)

Note: If it is given that (A, R) is a POSET, then 'A' is a set, and 'R' is a partial order relation on set A.

Slide

Note: Let 'A' is any set of real numbers, then $(A, \leq)$ is a POSET

↳ We are trying to say that relation "$\leq$" is a partial order relation on any set of real numbers

We know,
$(a,b) \in R$ iff $a \, R \, b$

$\therefore (a,b) \in \leq$ iff $a \leq b$

Let $A = \{-1, 2.5, 3\}$

$-1 \leq -1$ is true
$\therefore (-1,-1) \in \leq$

$\leq = \{ (-1,-1), (-1,2.5), (-1,3), (2.5,-1), (2.5,2.5), (2.5,3), (3,-1), (3,2.5), (3,3) \}$

$2.5 \leq -1$ is false
$\therefore (2.5,-1) \notin \leq$

It is Reflexive + Anti-symmetric + Transitive
$\therefore$ '$\leq$' on set A is partial order rel^n

Note: Let 'A' is any set of Positive integers $\because 0 \notin$ Set A then $(A, \div)$ is a POSET {where relation '$\div$' denotes "divides"} (don't confuse with "divided by")

$(a,b) \in \div$ (divides) iff 'a divides b' i.e. $a \div b$ is true

① Every positive integer divides itself $\because (a,a) \in \div, \forall a \in A$
Hence, Reflexive

② Let $a, b \in A$ such that $a \neq b$, then either $a \div b$ or $b \div a$ or 'none divides none'
but both can not be true i.e.
either $(a,b) \in \div$ & $(b,a) \notin \div$
or $(a,b) \notin \div$ & $(b,a) \in \div$
or $(a,b) \notin \div$ & $(b,a) \notin \div$
All three cases are valid w.r.t. Anti-symmetric
Hence, Anti-symmetric

③ If 'a divides b' & 'b divides c' then 'a divides c'
i.e. if $(a,b) \in \div$ & $(b,c) \in \div$, then $(a,c) \in \div \ \forall a,b,c \in A$
Hence, transitive

$\because$ on any set A of Positive integers, relation 'divides' is a partial order relation, and hence $(A, \div)$ is a POSET

Q:- Let 'C' is any Collection of sets and relation '$\subseteq$' denotes "Subset" or "Set Containment" Check whether $(C, \subseteq)$ is a POSET or not ?

Note:

Let $C = \{S_1, S_2, \ldots\}$ where $S_1, S_2, \ldots$ etc are sets.

Let C is any collection of sets, then $(C, \subseteq)$ is a POSET.

Let $S_1, S_2 \in C$
$(S_1, S_2) \in \subseteq$
if and only if
$S_1 \subseteq S_2$

① Every set is a subset of itself.
∴ $\forall S \in C, (S, S) \in \subseteq$, and hence Reflexive

② Let $S_1, S_2 \in C$, such that $S_1 \neq S_2$, then either $S_1 \nsubseteq S_2$ or $S_2 \nsubseteq S_1$ or ($S_1 \nsubseteq S_2$ and $S_2 \nsubseteq S_1$)
↳ Hence anti-symmetric

③ If $S_1 \subseteq S_2$ & $S_2 \subseteq S_3$, then $S_1 \subseteq S_3$, Hence Transitive



Topic : Comparability

- Let (A, R) be a POSET.
- Let $a, b \in A$,

  Elements 'a' and 'b' are said to be Comparable if and only if, either $a \, R \, b$ or $b \, R \, a$

  i.e. either $(a, b) \in R$ or $(b, a) \in R$

Let $a, b \in A$,

If neither $(a, b) \in R$ nor $(b, a) \in R$, then a & b are not Comparable w.r.t. POSET (A, R)

Slide

- In a POSET (A, R),

  R is a partial order relation,

  $\therefore$ R is guaranteed to be reflexive

and hence $\forall a \in A$,

$$(a, a) \in R$$

i.e. Every element is definitely comparable with itself.

eg: Let $A = \{1, 2, 3, 4\}$
and we know $(A, \div)$ is a POSET.

① '1' divides Every element of set A, i.e. '1' is Comparable with all elements of set A.

② '2' is Comparable with '1', '2' & '4'.
but '2' is not Comparable with '3',
because neither 2 divides 3 nor 3 divides 2
i.e. neither $(2,3) \in \div$ nor $(3,2) \in \div$

③ '3' is Comparable with '1' & '3'.
but '3' is not Comparable with '2' & '4'

④ '4' is Comparable with '1', '2' & '4'.
but not Comparable with '3'.

Note:

In a POSET (A, R).

Every pair of elements of set A need not be Comparable



Topic : Totally Ordered Set / (TOSET)

- Let (A, R) be a POSET,

  If Every pair of elements of set A are comparable, then POSET (A, R) is called a TOSET and relation 'R' can be called a "Total order relation"

Slide

Note :- ① Every TOSET is also a POSET,
But Every POSET need not be a TOSET

Similarly, ② Every Total order relation is also a Partial order relation, but Every Partial Order relation need not be a Total order relation.

eg: Let $A_1 = \{1, 2, 3, 4\}$
and $(A_1, \div)$ is a POSET

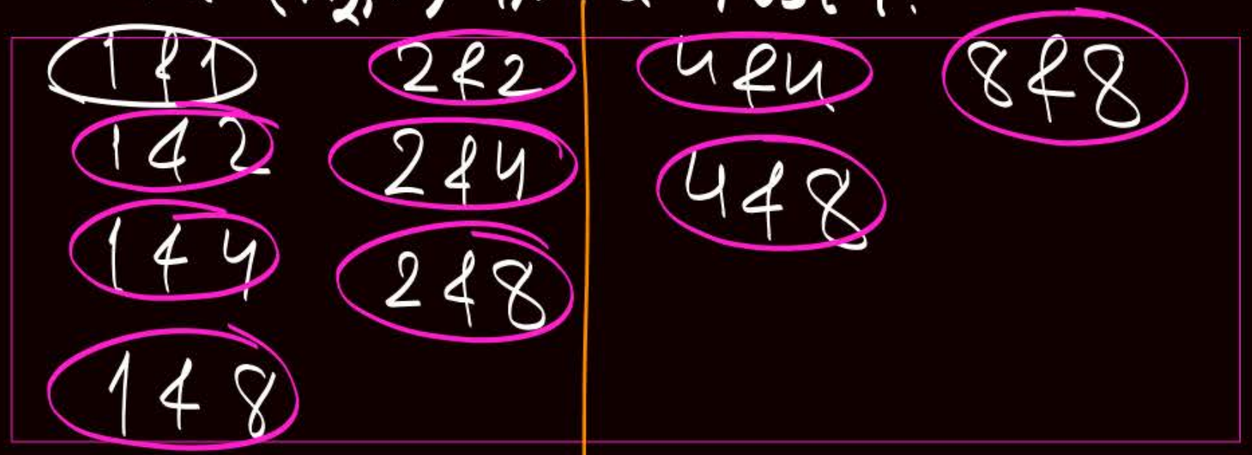
↓

There exists some pairs of elements in set A_1 that are not Comparable w.r.t. POSET $(A_1, \div)$ {eg 2 & 3}

↓

Hence, $(A_1, \div)$ is a POSET, but not a TOSET

eg: Let $A_2 = \{1, 2, 4, 8\}$
and $(A_2, \div)$ is a POSET.



In this POSET Every Pair of elements of set A_2 is Comparable,

∴ $(A_2, \div)$ is a TOSET

↳ Hence POSET as well



2 mins Summary

- Topic: Partial order relation
- Topic: Partially ordered set (POSET)
- Topic: Comparability
- Topic: Total order relation and TOSET

THANK - YOU